

AI for Bharat Hackathon **Green** Sentinel

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Protecting India's agriculture from **~₹2-3 lakh crore** annual losses due to pests, diseases, fires, and wildlife intrusions—leveraging AI for sustainable farming.

The Digital Immune System for Indian Agriculture

Green Sentinel is a production-deployed, cloud-native agricultural intelligence platform built for the realities of Bharat: Mobile-first (375px viewport), Offline-first (PWA), and WhatsApp-first.



Satellite Analytics

Weekly crop health scores without field visits via AI-powered analysis.



Edge AI Vision

Autonomous threat detection via Claude 3.5 Sonnet vision analysis.



AI Disease Forecasting

Predicting and diagnosing crop diseases in under 8 seconds.

World-Class Agricultural **AI**, Engineered for **Bharat's Constraints**

Why AI?

It replaces the expensive agronomist. Claude 3.5 Sonnet analyzes threats (fire, human intruders, nilgai) from cheap CCTV in seconds, bypassing the need for constant manual monitoring.

How AWS?

100% Serverless architecture. Powered by Lambda, DynamoDB, Amazon Bedrock, and API Gateway. We utilize a monolithic inline Lambda router with $O(1)$ path-based dispatch to minimize cold starts and keep infrastructure pay-per-request.

The Value Added

Immediate action despite language barriers. Delivers instant WhatsApp text alerts with threat snapshots to farmers' phones.

Comprehensive Intelligence Pipeline



Satellite Health Tracking

NDVI, NDWI, and Moisture Index calculations. AI-powered crop health scoring and trend analysis.



Weather Intelligence

Hyper-local, hourly conditions and 14-day forecasting via Open-Meteo.



Disease & Pest Forecasting

AI rule-based prediction alerting on Late Blight, Fall Armyworm, and Rust based on localized weather and crop stages.



Smart Irrigation Modeling

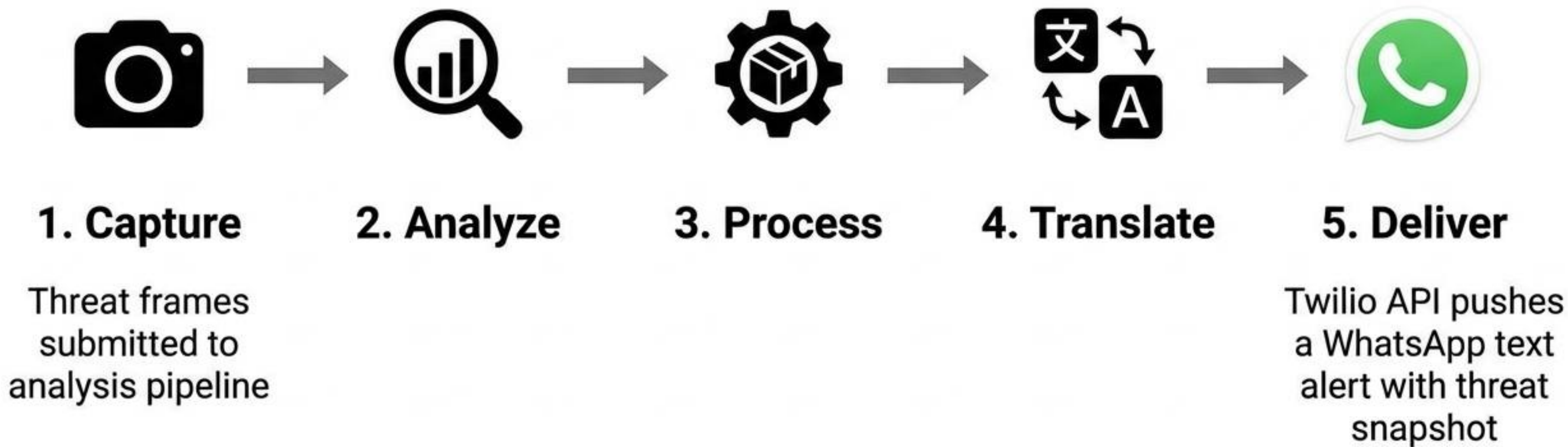
Zone-based scheduling derived from Evapotranspiration (ET_0) and soil moisture estimates.



Yield Prediction

Machine learning models factoring historical data and current vegetation indices.

The <10 Second End-to-End Alert Pipeline



UX Philosophy: Built for the 375px Viewport

Our interface is a zero-install Progressive Web App (PWA) built specifically for budget Androids on fluctuating 2G/3G networks.



Offline-First Architecture: Zustand stores persist to localStorage; Service Workers cache the UI and farm data.



Adaptive Complexity: A toggle between Simple Mode (large icons, basic metrics) and Expert Mode (detailed charts and health trends) accommodates different literacy levels.



Cloud-Native Serverless Topology



Frontend: CloudFront CDN hosting the React 18 / Vite PWA.



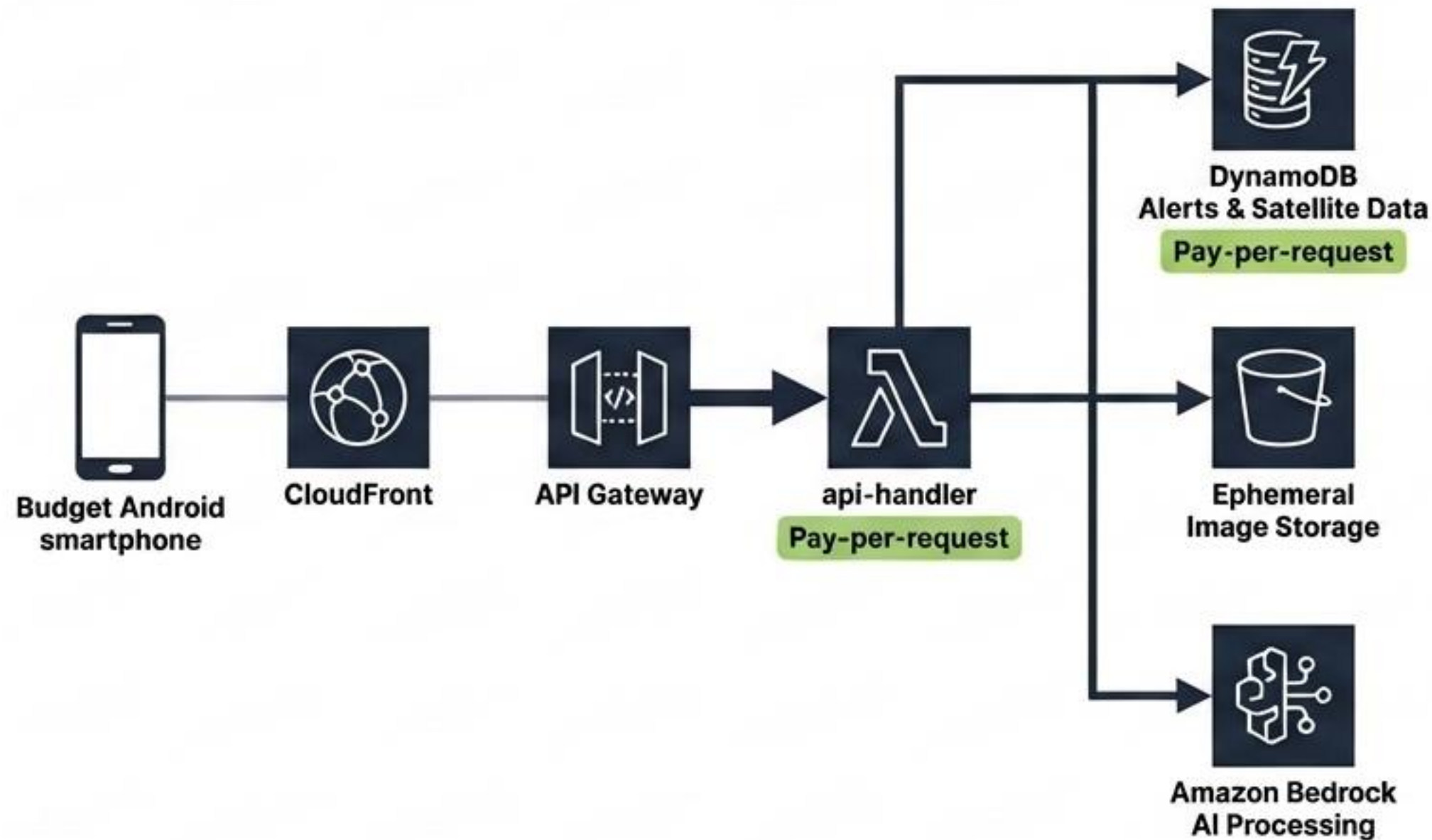
Compute: Monolithic api-handler Lambda in Node.js 20.x. Deliberate choice: one cold start pool, 8.7 KB inline code, zero zip dependencies.



Storage & State: DynamoDB using PAY_PER_REQUEST single-table design for alerts and satellite data. S3 for frame storage (24-hour lifecycle).



Orchestration: EventBridge triggering health analysis and alert pipelines.



The Technology Stack

Frontend Layer

React 18, Vite, TypeScript 5.3, Tailwind CSS, Zustand (persisted state), Workbox PWA.



Backend & AWS Layer

Node.js 20.x, AWS SDK v3, AWS CDK (Infrastructure as Code), AWS Cognito (Phone OTP Auth).



AI & Intelligence

Amazon Bedrock (Claude 3.5 Sonnet Vision)



External Data APIs

Open-Meteo (Free weather, 10K requests/day), Twilio (WhatsApp Sandbox)



Radical Affordability: ~₹730/Month for a 3-Farm Operation

Every infrastructure rupee is justified through strict AWS Free Tier compliance and AI budget guards.

Compute: Lambda & Cognito (1M+ requests)	Free Tier
Database: DynamoDB (~50K ops)	~₹20
Storage & CDN: S3 & CloudFront (10GB egress)	~₹110
AI Vision & Alerts: Bedrock Threat Detection (Claude 3.5 Sonnet calls)	~₹230 (\$1.80)

₹730/month

Strict Cost Guards: Server-side DynamoDB counters cut off API access after 200 daily AI calls to prevent runaway costs.

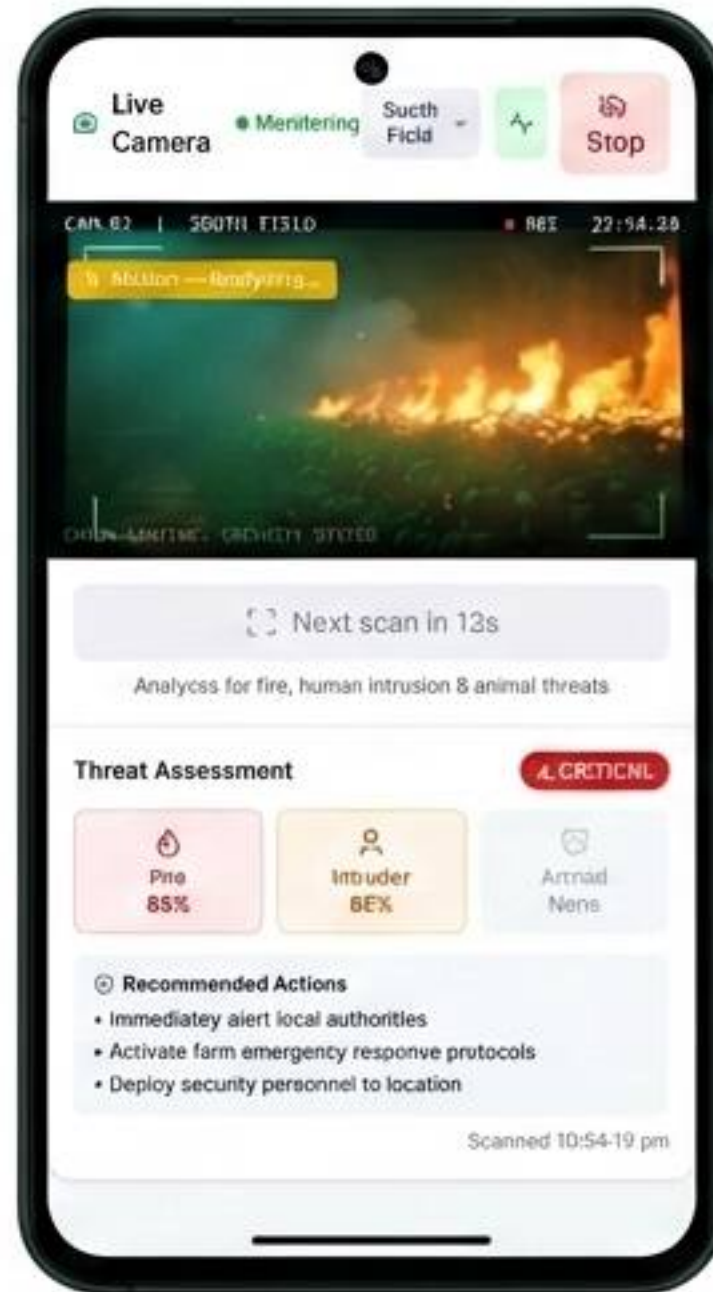
Prototype in Action: Threat Detection & Crop Health

AI threat detection with mock health scoring

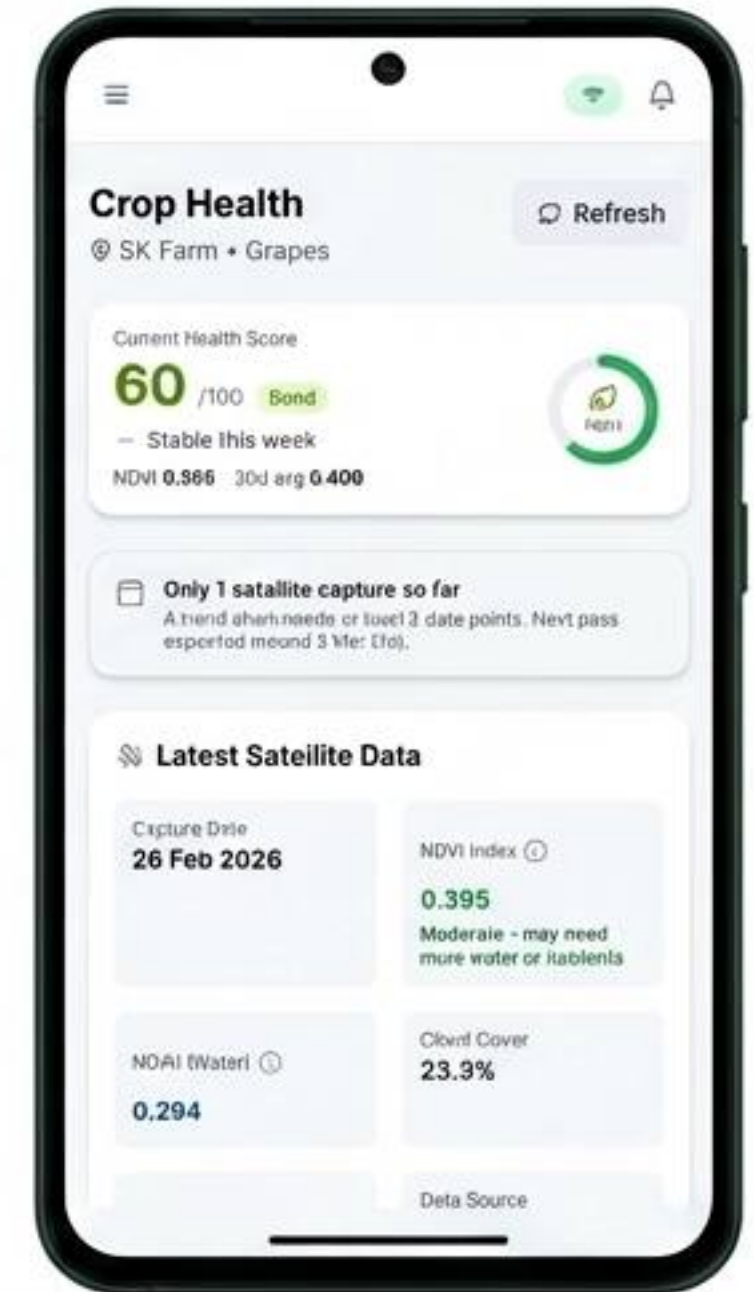
Critical Alerting: Identifying fires and human intruders with exact timestamps.



Live AI Camera: Threat detection powered by Claude 3.5 Sonnet Vision.



Satellite Crop Health: AI-powered health scoring with easy-to-read NDVI-style metrics.



Performance & Benchmarking

Engineered for the reality of rural network latency and critical threat timelines.

<10s

End-to-End Latency

From threat detection to WhatsApp alert delivery on the farmer's phone.

<3s

Frame Analysis

Amazon Bedrock Claude 3.5 Sonnet processing speed per image.

8s

PWA Load Time

Cold start initialization on standard 2G/3G networks.

99.5%

Target Uptime

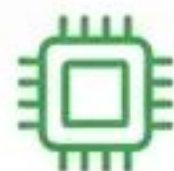
Supported by exponential backoff retries on Lambda functions and S3 temporary caching.

The 3 Moats & Future Scale

Our Competitive Moats:



1. The **Data Flywheel**: Every scan and alert improves our **localized** Indian disease models (e.g., Late Blight on Nashik grapes).



2. **Benign Hardware Lock-in**: Once Claude threat detection is deployed, **switching costs** are physical, not just digital.



3. **Deep Localization**: Bypasses **generic English AI chatbots** by automatically 'seeing' the **crop** and providing **threat intelligence**.



Prototype Assets



GitHub Public Repository

Access our complete AWS CDK infrastructure, React frontend, and Edge Agent source code.

<https://github.com/ShraddhaS1112/GreenSentinel/tree/main>



Demo Video

A 3-minute comprehensive walkthrough of the Green Sentinel Digital Immune System in action.

<https://github.com/ShraddhaS1112/GreenSentinel/blob/main/assets/demo/GreenSentinel.mp4>

Green Sentinel

Empowering Bharat's farmers with the cloud.

